Hallam Land Management, Taylor Wimpey & Hannick Homes

Eastern Villages, Swindon

Breeding and Winter Bird Survey Report

July 2013

FPCR Environment and Design Ltd
Registered Office: Lockington Hall, Lockington, Derby DE74 2RH
Company No. 07128076. [T] 01509 672772 [F] 01509 674565 [E] mail@fpcr.co.uk
[W] www.fpcr.co.uk

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1.0 INTRODUCTION

1.1 The following report has been prepared by FPCR Environment & Design Ltd on behalf of a consortium comprising Hallam Land Management Ltd, Taylor Wimpey and Hannick homes. It provides details of a series of breeding and wintering bird surveys undertaken during 2011, 2012 and 2013 on land at the eastern villages development area in Swindon, Wiltshire.

Site Location and Context

1.2 The Study Area is centred on the Ordnance Survey Grid Reference SU 2007 8752 and lies to the east of Swindon close to the village of South Marston.

1.3 The site is primarily used for agriculture at present, with pastoral fields bisected by hedgerow field boundaries in the west, to the south of South Marston, and a range of agricultural uses centred around Rowborough Farm to the east, including arable use, turf production and biomass fuel production.

1.4 The site and wider area to the south has been the focus of a wide range of surveys from at least 1999, including a number completed by FPCR. These surveys included bird surveys over a number of years both within the site and wider Eastern Villages Allocated site area.

Survey Objectives

1.5 To identify areas of ornithological interest and make recommendations to minimise the potential impact of development and where feasible to consider opportunities for additional habitat creation. The objectives of the survey were to:

- Identify the presence and distribution of wintering birds on the site;
- Evaluate the importance of local bird populations and their habitat requirements;
- Evaluate the conservation importance of the site;
- To identify areas of ornithological interest and make recommendations to minimise the potential impact of development and where feasible to consider opportunities for additional habitat creation.

2.0 LEGISLATION

2.1 The Wildlife and Countryside Act 1981 (as amended) is the principal legislation affording protection to UK wild birds. Under this legislation all birds, their nests and eggs are protected by law and it is an offence, with certain exceptions to recklessly or intentionally:

- Kill, injure or take any wild bird;
- Take, damage or destroy the nest of any wild bird while in use or being built;
- Take or destroy the egg of any wild bird.

2.2 Species listed on Schedule 1 of the Wildlife and Countryside Act 1981 (as amended) are specially protected at all times.

2.3 A number of birds feature on the Natural Environment and Rural Communities (NERC) Act, Section 41 (S41) as species which are of principal importance for the conservation of biodiversity in England. The S41 list is used to guide decision-makers such as public bodies, including local and regional authorities, in implementing their duty under section 40 of the NERC Act, to have regard to the conservation of biodiversity in England, when carrying out their normal functions.

2.4 In addition to statutory protection, some bird species are classified according to their conservation status, such as their inclusion on the Red and Amber lists of Birds of Conservation Concern (BoCC) in the UK (Eaton et al 2009)[1]:

- Red list (high conservation concern) species are those that are Globally Threatened according to IUCN criteria; those whose population has declined rapidly (50% or more) in recent years; and those that have declined historically and not shown a substantial recent recovery.
- Amber list (medium conservation concern) species are those with an unfavourable conservation status in Europe; those whose population or range has declined moderately (between 25% and 49%) in recent years; those whose population has declined historically but made a substantial recent recovery; rare breeders; and those with internationally important or localised populations.
- Green list (low conservation concern) species fulfil none of the above criteria.

2.5 The site falls within the area covered by the Wiltshire Local Biodiversity Action Plan (LBAP). A number of bird species are covered within this BAP under a number of different Habitat Action Plans (HAPs). Relevant HAPs on site would include; built environment, rivers and streams, farmland, woodland and hedgerows.

3.0 METHODOLOGY

Breeding Bird Survey

3.1 The survey methodology employed was broadly based on that of territory mapping[2] as used for the British Trust for Ornithology (BTO) Common Bird Census (CBC). Standard BTO species codes and symbols for bird activities were used to identify birds and denote activity, sex and age where appropriate. The criteria used in the assessment of breeding birds has been adapted from the standard criteria proposed by the European Ornithological Atlas Committee (EOAC 1979) and are grouped into three categories: possible breeder e.g. birds observed in suitable habitat or singing male recorded, probable breeder e.g. pair in suitable habitat, territory defended, agitated behaviour or nest building and confirmed breeder e.g. recently fledged young observed, adult birds carrying food for young. Birds that were considered to be not using the site for breeding were categorised as ónon breedersôe.g. flying over the site, migrant, habitat not suitable.

[1] Eaton, M.A. *et al*. 2009. Birds of Conservation Concern 3: the population status of birds in the United Kingdom, Channel Islands and Isle of Man. *British Birds* 102:296-341
[2] Bibby *et al*. 1992

3.2 To provide a reasonable level of accuracy for determining the population status of the breeding birds on the site, three surveys were undertaken between 05.00 and 11.00 during the period May and June 2011. A route was mapped out prior to the surveys being undertaken, paying particular attention to any linear features, such as hedgerows and tree lines, and natural features such as ponds, lakes, areas of scrub and woodland. Bird surveys were not undertaken in unfavourable conditions such as heavy rain or strong wind, which may negatively affect the results. Table 1 provides details of the survey dates and weather conditions.

Table 1. Breeding Bird Survey Dates and Conditions

Date	Cloud Cover	Rain	Wind	Visibility
12/05/11	20%	None	Gentle Breeze	Good
01/06/11	40%	None	Light Air	Good
21/06/11	70%	None	Gentle Breeze	Excellent

Wintering Bird Survey

3.3 A total of three winter bird surveys were undertaken during the winter period of 2012/13 to determine winter bird use and assess the sites importance to local bird populations. Particular attention focused on the presence of over wintering or passage waders and passerines. Standard BTO species codes and symbols for bird activities were used as with the breeding bird surveys. Table 2 provides details of the surveys dates and weather conditions.

Table 2. Wintering Bird Survey Dates and Conditions

Date	Cloud Cover	Rain	Wind	Visibility
18/12/12	30%	1 Light Shower	Light Air	Excellent
10/01/13	100%	Light and Occasional	Light Air	Good
04/02/13	70%	None	Gentle Breeze	Excellent

Survey Area

3.4 The area surveyed during both the breeding and wintering bird surveys varied slightly, conforming to the emerging development framework plan at the time of survey. On each occasion, the area surveyed covered at least that forming the current application site boundary (as of June 2013). The survey areas for the breeding and wintering bird surveys are shown on Figures 1 and 2 respectively.

Previous Bird Surveys

3.5 A series of both breeding and wintering bird surveys were undertaken in 2006 and in to 2007 by FPCR. A summary of those notable species recorded (i.e. those listed on

Schedule 1 of the Wildlife and Countryside Act 1981 (as amended), species of principal importance under S.41 of the NERC Act and those listed as BoCC amber and red listed species of conservation concern) are presented within this report.

Assessment

3.6 The conservation value of bird populations has been measured using two separate approaches: nature conservation value and conservation status. The IEEM guidance on ecological impact assessment assesses nature conservation value within a geographical context. To attain each level of value, an ornithological resource or one of the features (species population or assemblage of species) should meet the criteria set out in Table 3. In some cases, professional judgement may be required to increase or decrease the allocation of specific value, based upon local knowledge.

3.7 The recent county annual bird report, the Wiltshire Bird Report 2010 (Hobby 37) as published by the Wiltshire Ornithological Society was consulted to inform the assessment.

Table 3. Definition of Terms Relating to Nature Conservation Value

Nature Conservation Value	Examples of Selection Criteria
International	A species which is part of the cited interest of an SPA and which regularly occurs in internationally or nationally important numbers. A species present in internationally important numbers (>1% of international population).
National	A species which is part of the cited interest of a SSSI and which regularly occurs in nationally or regionally important numbers. A nationally important assemblage of breeding or over-wintering species. A species present in nationally important numbers (>1% UK population). Rare breeding species (<300 breeding pairs in the UK).
Regional	Species of principle importance under S41 of the NERC Act, which are not covered above, and which regularly occur in regionally important numbers. Species present in regionally important numbers (>1% of regional population). Sustainable populations of rare or scarce species within a region. Species on the BoCC Red List and which regularly occurs in regionally important numbers.

Nature Conservation Value	Examples of Selection Criteria
County	Species of principle importance under S41 of the NERC Act, which are not covered above, and which regularly occur in county important numbers. Species present in county important numbers (>1% of county population). Sustainable populations of rare or scarce species within a county or listed in a county BAP. A site designated for its county important assemblage of birds (e.g. a SINC Site). Species on the BoCC Red List and which regularly occur in county important numbers.
District	Species of principle importance under S41 of the NERC Act, which are not covered above, and are rare in the locality or in the relevant Natural Area profile. Species present in numbers just short of county importance. Sustainable populations of rare or scarce species within the locality. A site whose designation falls just short for inclusion for its county important assemblage of birds (e.g. a SINC Site). Other species on the BoCC Red List and which are considered to regularly occur in district important numbers.
Local	Other species of conservation interest (e.g. all other species of principle importance under S41 of the NERC Act and on the BoCC Red and Amber lists which are not covered above) regularly occurring in locally sustainable populations.
Site	All other BoCC Green-listed common and widespread species.

4.0 RESULTS

Previous Breeding and Wintering Bird Surveys

4.1 A total of 33 notable bird species were recorded during both the breeding and wintering bird surveys undertaken in 2006 and 2007, 18 and 26 of which were recorded in the breeding and wintering periods respectively. A summary of those notable species recorded is included within table 4.

Table 4. Notable Species Recorded during Breeding and Wintering Surveys 2006/07

Species	Legal/ Conservation status	Breeding Bird Surveys 2006	Wintering Bird Surveys 2006/07
Black-headed Gull *Chroicocephalus ridibundus*	Amber List		✓
Bullfinch *Pyrrhula pyrrhula*	Red List NERC S.41 LBAP	✓	✓
Dunnock *Prunella modularis*	Amber List NERC S.41	✓	✓
Fieldfare *Turdus pilaris*	Schedule 1 Red List		✓
Grasshopper Warbler *Locustella naevia*	Red List NERC S.41 LBAP	✓	
Green Woodpecker *Picus viridis*	Amber List		✓
Grey Wagtail *Motacilla cinerea*	Amber List		✓
Greylag Goose *Anser anser*	Amber List		✓
Herring Gull *Larus argentatus*	Red List NERC S.41 LBAP		✓
House Martin *Delichon urbica*	Amber List LBAP	✓	
House Sparrow *Passer domesticus*	Red List NERC S.41 LBAP	✓	✓
Kestrel *Falco tinnunculus*	Amber List		✓
Lapwing *Vanellus vanellus*	Red List NERC S.41 LBAP		✓
Lesser Black-baked Gull *Larus fuscus*	Amber List		✓

Species	Legal/ Conservation status	Breeding Bird Surveys 2006	Wintering Bird Surveys 2006/07
Linnet *Carduelis cannabina*	Red List NERC S.41 LBAP	✓	✓
Mallard *Anas platyrhynchos*	Amber List	✓	✓
Meadow Pipit *Anthus pratensis*	Amber List		✓
Merlin *Falco columbarius*	Schedule 1 Amber List LBAP		✓
Mistle Thrush *Turdus viscivorus*	Amber List	✓	✓
Red Kite *Mivus milvus*	Schedule 1 Amber List		✓
Redwing *Turdus iliacus*	Schedule 1 Red List		✓
Reed Bunting *Emberiza schoeniclus*	Amber List NERC S.41 LBAP	✓	✓
Skylark *Aluda arvensis*	Red List NERC S.41 LBAP	✓	✓
Snipe *Gallinago gallinago*	Amber List LBAP		✓
Song Thrush *Turdus philomelos*	Red List NERC S.41 LBAP	✓	✓
Starling *Sturnus vulgaris*	Red List NERC S.41 LBAP	✓	✓
Swallow *Hirundo rustica*	Amber List LBAP	✓	
Swift *Apus apus*	Amber List LBAP	✓	
Teal *Anas crecca*	Amber List		✓
Whitethroat *Sylvia communis*	Amber List	✓	
Willow Warbler *Phylloscopus trochilus*	Amber List	✓	
Yellow Wagtail *Motacilla flava*	Red List NERC S.41 LBAP	✓	
Yellowhammer *Emberiza citrinella*	Red List NERC S.41 LBAP	✓	✓

Breeding Bird Survey 2011

Summary

4.2 A total of 56 bird species were recorded during the breeding bird surveys within the site. Of the species observed, 24 are either protected, appear on the RSPB BoCC as declining (red or amber lists) or are listed as Species of Principal Importance. A full list of results can be found in Appendix A, whilst the recorded locations of the 24 notable species can be found on Figure 1 and detailed in Table 5.

4.3 Of the 56 species recorded, 13 were confirmed as breeding including yellow wagtail *Motacilla flava*, yellowhammer *Emberiza citronella* (BoCC red list, S.41, LBAP), swallow *Hirundo rustica* (BoCC amber list, LBAP) and whitethroat *Sylvia communis* (BoCC amber list). A further 22 species were considered probable breeders, whilst the remaining 21 species were considered possible breeders or non-breeders.

Table 5. Species of Principal Importance, BoCC Red and Amber Listed Bird Species Recorded at East Swindon during Breeding Bird Surveys 2011, and their Recent Status within Wiltshire.

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Breeding Status†	Recent Status in Wiltshire
Bullfinch *Pyrrhula pyrrhula*	Amber List NERC S.41 LBAP	2 / 3	Probable	Moderately common resident of woods and scrub
Cuckoo *Cuculus canorus*	Red List NERC S.41 LBAP	1 / 1	Possible	Widespread but decreasing visitor from Africa
Dunnock *Prunella modularis*	Amber List NERC S.41	10 / 3	Probable	Still common and widespread resident, though decreasing Britain as a whole
Green Woodpecker *Picus viridis*	Amber List	1 / 2	Possible	Widespread resident, increasing Britain as a whole
Grey Partridge *Perdix perdix*	Red List NERC S.41 LBAP	2 / 1	Possible	Uncommon and Long decreasing resident, some released/naturalised stock
Herring Gull *Larus argentatus*	Red List NERC S.41 LBAP	2 / 2	Non-breeder	Scarce visitor from elsewhere in Britain/north Europe. Recent and increasing local breeder
House Martin *Delichon urbica*	Amber List LBAP	24 / 2	Possible	Common summer visitor/passage
House Sparrow *Passer domesticus*	Red List NERC S.41 LBAP	4 Colonies / 3	Probable	Common resident, trends in Wiltshire uncertain but large decrease nationally since 1970s
Kestrel *Falco tinunculus*	Amber List	1 / 1	Possible	Common but decreasing resident, also passage migrant and winter visitor
Lesser Black-backed Gull *Larus fuscus*	Amber List	3 / 1	Non-breeder	Increasing passage/winter from GB/Iceland/north Europe, recent and increasing local breeder

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Breeding Status†	Recent Status in Wiltshire
Linnet *Carduelis cannabina*	Red List NERC S.41 LBAP	8 / 3	Probable	Seriously decreasing resident, also passage migrant and winter visitor
Mistle Thrush *Turdus viscivorous*	Amber List	4 / 2	Probable	Common resident and perhaps partial migrant
Red Kite *Milvus milvus*	Schedule 1 Amber List	1 / 1	Non-breeder	Rare resident and scarce wanderer from Wales and re-established populations in England
Reed Bunting *Emberiza schoeniclus*	Amber List NERC S.41 LBAP	5 / 2	Probable	Seriously decreasing resident in wetlands, also some passage/winter
Skylark *Alauda arvensis*	Red List NERC S.41 LBAP	5 / 3	Probable	Still locally common resident/passage/winter, but serious declines in GB
Song Thrush *Turdus philomelos*	Red List NERC S.41 LBAP	5 / 3	Probable	Common resident, partial migrant, also winterer from north GB/Europe
Starling *Sturnus vulgaris*	Red List NERC S.41 LBAP	55 / 3	Probable	Seriously decreasing resident, common winter visitor from north Europe
Stock Dove *Columba oenas*	Amber List	7 / 3	Probable	Common and increasing resident of parkland, downland, wood edges, farmland with trees and barns, locally towns
Swallow *Hirundo rustica*	Amber List LBAP	47 / 3	Confirmed	Common and increasing summer/passage
Swift *Apus apus*	Amber List LBAP	5 / 2	Non-breeder	Common summer visitor
Whitethroat *Sylvia communis*	Amber List	24 / 2	Confirmed	Common summer visitor and passage migrant
Willow Warbler *Phylloscopus trochilus*	Amber List	15 /3	Probable	Common summer visitor
Yellow Wagtail *Motacilla flava*	Red List NERC S.41 LBAP	4 / 3	Confirmed	Local summer visitor
Yellowhammer *Emberiza citronella*	Red List NERC S.41 LBAP	9 / 3	Confirmed	Decreasing resident, though still widespread on arable farmland

Description

4.4 The majority of bird species recorded are typical of the range of habitats which predominate within the site being heavily influenced by the current agricultural management of the site. The Site is dominated by arable fields (particularly centrally) and improved pasture bisected by hedgerows. These habitats attracted a range of common and widespread, although declining farmland bird species including linnet, skylark and yellowhammer in small numbers, all of which are BoCC red list, S.41, and

listed on the LBAP. The latter was confirmed to be breeding on the edge of a large arable field compartment to the east of Rowborough farm. This field compartment and associated hedgerows as indicated by TN1 on figure 1 was considered to be of greatest value to farmland birds. In addition to the aforementioned farmland species, yellow wagtail, a species generally occurring more localised was recorded within the field compartment and confirmed to be breeding within the arable crop at the time of survey. During 2013, this field compartment was observed to be managed for the purpose of growing and supplying amenity grassland (turf) and it is therefore likely that its value to farmland birds would have recently reduced.

4.5 The large number of hedgerows present within the site are managed in a way which offer good opportunities to breeding birds. Hedgerow, mature tree standards and scrub habitats throughout the site attracted a number of notable species often associated with woodland edge habitat including small numbers of bullfinch, dunnock, song thrush and stock dove. Larger numbers of those notable warblers associated with scrub habitat including whitethroat and willow warbler were also scattered throughout the site. In addition to these notable species, the hedgerows attracted a number of common finches, thrushes and other warbler species.

4.6 Reed bunting, a declining resident species and listed on the BoCC amber list and LBAP was recorded locally within the east of the site associated with the River Cole corridor.

4.7 The assemblage recorded is representative of that recorded during previous breeding bird surveys in 2006 in terms of the notable species observed. Surveys from both years indicate that the site is occasionally visited by other species considered to occur more locally or are declining or uncommon. Such species include grey partridge, cuckoo, red kite and grasshopper warbler. In the case of red kite, only one bird was observed to fly over the site on one occasion.

Breeding Bird Assemblage Value

4.8 The surveys have highlighted that the site supports a number of notable species including a range of declining farmland birds. However, all are typical of the habitats present on site comprising predominantly heavily managed arable habitats which are a common feature of the surrounding landscape. As no significant populations of any notable species have been recorded, the site is considered of Local nature conservation value.

Wintering Bird Surveys 2012/13

Summary

4.9 A total of 54 bird species were recorded during the wintering bird surveys within the site. Of the species observed, 28 are either protected, appear on the RSPB BoCC as declining (red or amber lists) or are listed as Species of Principal Importance. A full list of results can be found in Appendix B, whilst the recorded locations of the 10 notable species can be found on figure 2 and detailed in table 6.

Table 6: Species of Principal Importance, BoCC Red and Amber Listed Bird Species Recorded at East Swindon during Wintering Bird Surveys 2012/13, and their Recent Status within Wiltshire.

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Recent Status in Wiltshire
Barn Owl *Tyto alba*	Schedule 1 Amber List LBAP	1 / 1	Scarce but widespread resident, increasing again after serious decline
Black-headed Gull *Larus ridibundus*	Amber List	56 / 2	Local breeder, common winter/passage
Bullfinch *Pyrrhula pyrrhula*	NERC S.41 Amber List LBAP	6 / 3	Moderately common resident of woods and scrub, visiting gardens in winter
Dunnock *Prunella modularis*	NERC S.41 Amber List	18 /3	Common and widespread resident, decreasing in Britain
Fieldfare *Turdus pilaris*	Schedule 1 Red List	68 / 3	Common winter/passage
Great Black-backed Gull *Larus marinus*	Amber List	2 / 1	Scarce migrant/winter visitor, breeds coastal and north Europe, winters also inland in GB
Green Sandpiper *Tringa ochropus*	Schedule 1 Amber List	1 / 1	Uncommon passage migrant and winter visitor
Green Woodpecker *Picus viridis*	Amber List	2 / 1	Widespread resident, increasing throughout Britain
Grey Wagtail *Motacilla cinerea*	Amber List	1 / 1	Common winter visitor, also some passage/winter from north Britain
Herring Gull *Larus argentatus*	NERC S.41 Red List LBAP	70 / 3	Scarce winter visitor from elsewhere in Britain/north Europe, recent and increasing local breeder
House Sparrow *Passer domesticus*	NERC S.41 Red List LBAP	5 Colonies / 3	Common resident, trends in Wiltshire uncertain but decreasing nationally
Kestrel *Falco tinunculus*	Amber List	2 / 1	Common but decreasing resident, also passage migrant and winter visitor
Lesser Black-backed Gull *Larus fuscus*	Amber List	190 / 3	Increasing passage/winter from GB/Iceland/ north Europe, recent and increasing local breeder
Lesser Redpoll *Carduelis cabaret*	NERC S.41 Red List LBAP	4 / 1	Scarce winter visitor from north England, no recent evidence of breeding
Linnet *Carduelis cannabina*	NERC S.41 Red List LBAP	6 / 3	Seriously declining resident, also passage migrant and winter visitor
Little Egret *Egretta garzetta*	Amber List	1 / 1	Resident, but localised breeder
Mallard *Anas platyrhynchos*	Amber List	7 / 3	Common resident (some released) and winter visitor from Europe

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Recent Status in Wiltshire
Meadow Pipit *Anthus pratensis*	Amber List	6 / 3	Common resident, and passage/winter
Mistle Thrush *Turdus viscivorous*	Amber List	5 / 2	Common resident and perhaps partial migrant
Red Kite *Milvus milvus*	Schedule 1 Amber List	1 / 1	Rare resident and scarce wanderer from Wales and re-established populations in England
Redwing *Turdus iliacus*	Schedule 1 Red List LBAP	153 / 3	Common winter/passage
Reed Bunting *Emberiza schoeniclus*	NERC S.41 Amber List LBAP	12 / 3	Seriously decreasing resident in wetlands, also some passage/winter
Skylark *Alauda arvensis*	NERC S.41 Red List LBAP	33 / 3	Locally common resident/passage/winter, but serious decline in GB
Song Thrush *Turdus philomelos*	NERC S.41 Red List LBAP	9 / 3	Common resident, partial migrant, also winter from north GB/Europe
Starling *Sturnus vulgaris*	NERC S.41 Red List LBAP	189 / 3	Seriously declining resident, common winter visitor
Stock Dove *Columba oenas*	Amber List	1 / 1	Common and increasing resident
Woodcock *Scolopax rusticola*	Amber List	3 / 3	Decreasing resident, under recorded winter visitor from north/east Europe
Yellowhammer *Emberiza citronella*	NERC S.41 Red List LBAP	42 / 3	Decreasing resident, though still quite widespread on arable farms

Description

4.10 As per the breeding bird surveys, the site supported an assemblage of birds typical of the habitats available. Perhaps of greatest value was the farmland bird assemblage comprising small numbers of linnet and meadow pipit with moderate sized flocks of skylark and yellowhammer. The latter was consistently recorded in moderate numbers and within those arable field compartments supporting winter stubble.

4.11 Five Schedule 1 species were recorded including the common and widespread species, redwing and fieldfare. Barn owl, green sandpiper and red kite were all observed singularly. The barn owl was observed within the south east of the site hunting over grassland in the region where two barn owl boxes are located along the site boundary (TN1, Figure 2). It is possible that one of these is being used as a barn owl roost as the other was in a bad state of repair. An ephemeral and cattle poached water body within the west of the site provided foraging opportunities for green sandpiper. As with the observation of red kite during the breeding bird surveys, one was also observed to fly north over the site during the winter.

4.12 Reed buntings in flocks numbering up to 12 birds were again associated with the River Cole corridor within the east of the site.

4.13 Notable species associated with the hedgerows, woodland edge and scrub habitats included bullfinch, dunnock and song thrush, however all were recorded in relatively low numbers given the extent of suitable habitat available.

4.14 Results of recent wintering bird surveys and those undertaken in 2006/07 indicate that the site hosts a number of species associated with wetland habitats during winter including green sandpiper, grey wagtail, greylag goose, lapwing, little egret, mallard and snipe. In most part, these species were recorded occasionally and localised to the River Cole and associated inundated habitat, the South Marston Brook and a shallow ephemeral water body to the south of the site where the green sandpiper was recorded.

4.15 Flocks of mixed gulls were observed to fly over the site late in the afternoon on each survey occasion, and were considered to be moving to a roost.

Wintering Bird Assemblage Value

4.16 Surveys have identified that the site is used by a range of common and widespread species including a range of declining but still common farmland species. Occasional use of the site by more localised species was observed within more discreet habitat features such as the water bodies on site. However, no significant flocks of notable bird species were recorded and the site is considered to be of Local nature conservation value with emphasis on the farmland bird assemblage, notably yellowhammer.

5.0 DISCUSSION

Impacts of Habitat Loss to Breeding and Wintering Birds

5.1 The impact on breeding and wintering bird species arising from the potential effects of development is based upon an understanding of each speciesô ecological requirements, the type of development, number of birds recorded on site, their nature conservation criteria based on legislation and current guidance (e.g. Red and Amber listed Birds of Conservation Concern 3 (2009); Species listed as principal importance (S.41 NERC Act) and Local BAP species), their county status according to The Wiltshire Annual Bird Report 2010 (Hobby 37) and professional judgement.

5.2 The species recorded during the most recent surveys on site that are arguably the most vulnerable to impacts are the nineteen notable species that appear on Schedule 1 of the W&CA, the BoCC Red list and are listed as species of principal importance (Tables 5 & 6). The habitat requirements, species account, nature conservation value of these species are summarised in Table 7. In addition residual impacts associated with habitat loss have been assessed against the proposals as set out in the development framework plan (drawing number LHL005-126).

Development Proposals

5.3 The application boundary forming 162ha is to comprise new housing, infrastructure, proposed neighbourhood centre, education sites, outdoor sports facilities , allotments and community and local centres. Green infrastructure comprising formal recreational areas and less formal areas managed for the benefit of wildlife will stretch throughout the site linking up a number of existing and retained semi-natural habitats and will equate to almost 40% of the total site area.

Table 7. Evaluation of Species Considered Potentially Sensitive to Habitat Loss/Change

Species	Habitat Requirements	Species Account	Nature Conservation Value†	Characterisation of Unmitigated Impact	Suggested Possible Mitigation/ Compensation/ Enhancements	Residual Impact††
Barn Owl	Mainly associated with open but not treeless lowlands, especially farmland with spinneys, hedges, ditches, ponds, and banks, and some rough grass and roadside verges where mice and other mammal prey can be hunted in low flight.	Wintering ï A single barn owl was observed over the site in close proximity to existing barn owl boxes where a roost is possible	**Local**	Loss of hunting habitat in the form of disturbance free grassland and arable margins.	Although it is considered unlikely to compensate the overall loss of foraging habitat available on site, discreet areas of less disturbed grassland within the green infrastructure may continue to offer some hunting opportunities. The nest boxes should be retained in their current position and possibly replaced with new ones adding a layer of pellets from the old ones if present, as this will encourage continued use. Consideration should be given to the placement of other barn owl nest boxes where appropriate.	**Negligible**
Bullfinch	Breeds mainly in broad-leaved woods, but also commonly in dense hedgerows and thickets. Adults forage on seeds and vegetation, but invertebrates are important in diet of young.	Breeding ï Recorded within hedgerows to the east of the site in small numbers. Wintering ï Recorded scattered throughout the site in scrub, woodland edge and more established hedgerows.	**Local**	Loss of hedgerow sections which provide suitable foraging and nesting opportunities.	Wherever feasible, scrub habitat will be retained particularly along the boundaries of the site. Strategic planting should include native seed and fruit-bearing species such as ash, rowan and crab apple. Retained plantation woodland will continue to offer foraging and nesting opportunities in the short term prior to the establishment of new areas of planting.	**Negligible**

Species	Habitat Requirements	Species Account	Nature Conservation Value†	Characterisation of Unmitigated Impact	Suggested Possible Mitigation/ Compensation/ Enhancements	Residual Impact††
Cuckoo	Ranges over a wide variety of habitats including coniferous or broad-leaved woodland and scrub, parkland, open country with scattered trees or hedgerows including farmland of various types, wetlands with reedbeds, heaths, and costal dunes or marshes.	Breeding ï A single cuckoo was recorded along the River Cole corridor within a willow coppice which is outside of the application boundary.	**Local**	The intensively managed habitats comprising predominantly arable farmland do not offer good opportunities to this species. The bird recorded on site is most likely to be using those semi natural habitats along the river corridor which are not to be affected by proposals and buffered from development.	None suggested	**Negligible**
Dunnock	Commonly invades a wide variety of scrub grown situations. Has adapted to field hedgerows, farms, railway embankments, parks, gardens and vacant urban land. Feeds mainly on insects but small seeds are an important winter food.	Recorded mainly within hedgerows across the site in both breeding and wintering seasons.	**Local**	As with bullfinch, the loss of hedgerow sections which provide suitable foraging and nesting opportunities.	The inclusion new planting throughout the Green Infrastructure will continue to provide suitable conditions to support a large population of dunnock throughout the site. Dunnock will also use established habitats around the more formal areas of green infrastructure including parks and gardens.	**Negligible**

Species	Habitat Requirements	Species Account	Nature Conservation Value†	Characterisation of Unmitigated Impact	Suggested Possible Mitigation/ Compensation/ Enhancements	Residual Impact††
Fieldfare	Widespread winter visitor occurring almost anywhere. Feeds along hedgerows and in orchards, pastures and other areas of short grass, and on arable fields. Feeding sites are often close to woodland and tall hedges. Nomadic in winter as they travel the countryside for food, sometimes joined by other thrushes and starlings.	Wintering ï Recorded in small numbers throughout the site, mainly in hedgerows.	**Local**	Loss of foraging habitat in the form of open pasture fields, hedgerows and scrub habitat.	Provision of new planting should include the provision of fruiting species which will provide further foraging habitat for moderate numbers of fieldfare.	**Negligible**
Green Sandpiper	During the winter and on passage, found in ditches on lowland farmland, near streams, freshwater marshes, sewage works and watercress beds.	Wintering ï One bird located foraging within a cattle poached and shallow pool to the west of the site.	**Local**	Potential loss of some suitable water bodies which provide occasional foraging opportunities to overwintering and passage birds.	None suggested. Sections of the South Marston brook, on site ditches and the inclusion of other wetland features will continue to provide potential foraging habitat for occasional visits by green sandpipers.	**Negligible**
Grey Partridge	Breeds in lowlands, mostly in central and eastern England. Prefers open areas of low grass with dense cover.	Breeding ï Two birds were observed to the east of the site outside of the application site boundary.	**Local**	Loss of arable field compartments will reduce the overall available habitat to grey partridge on site; however it isnû clear to what extent the species uses the site. As no further observations were made, it is unlikely that the site is of significant value to the species.	None suggested	**Negligible**

Species	Habitat Requirements	Species Account	Nature Conservation Value†	Characterisation of Unmitigated Impact	Suggested Possible Mitigation/ Compensation/ Enhancements	Residual Impact††
Herring Gull	Most nest near the sea although there are a few inland colonies. During winter this species is associated with arable farmland which is used for foraging and loafing. Also feeds on urban rubbish tips, visits town parks and roost during the day on playing fields and at night on reservoirs and estuaries.	Breeding ï Two birds were observed to fly over the site only. Wintering ï Small to moderate numbers were observed to fly over the site during late afternoon along with other gull species and assumed to be heading to roost. Small numbers were recorded on site within the open arable fields.	**Site**	Loss of some potential winter foraging habitat in the form of arable fields.	None suggested. The inclusion of school playing fields and sports amenity fields within the proposals will continue to offer winter roosting areas.	**Negligible**
House Sparrow	Often associated with man, will avoid closed or dense vegetation, and except for seasonal foraging in corn fields and other crops, will usually avoid open terrain lacking in shrubs, trees and other cover.	Colonies were distributed within the west of the site in both seasons along the residential edge of South Marston and at the locations of the farms.	**Local**	Possible minor loss of some sections of hedgerows where colonies have been recorded.	House sparrows will readily habituate to new residential areas. Strategic placement of house sparrow nest-box terraces around the development will provide new breeding sites. The retention of boundary trees and hedgerows combined with new residential gardens and buildings will increase nesting and foraging opportunities.	**Positive**
Lesser Redpoll	Mainly distributed in open scrub woodland, often on hillsides and heaths and in field hedgerows, gardens, alder carrs along streams, and increasingly in young conifer plantations.	Wintering ï Four birds were observed on one occasion along the eastern boundary to the application site.	**Local**	Minor loss of potential foraging habitat though loss of tall herb patches and arable field margins.	The creation of informal habitat areas within the green infrastructure will continue to provide weedy areas which will offer foraging opportunities to small numbers of lesser redpoll.	**Negligible**

Species	Habitat Requirements	Species Account	Nature Conservation Value†	Characterisation of Unmitigated Impact	Suggested Possible Mitigation/ Compensation/ Enhancements	Residual Impact††
Linnet	Nests in areas of dense scrub or hedgerows. Feeds on small to medium sized seeds. Particularly dependent on weeds of open country and waste ground.	Breeding ï Small numbers recorded within the arable habitats on site and particularly centrally adjacent to Rowborough Farm (TN1, Figure 1) where breeding was considered probable. Wintering ï Again recorded within arable habitats around Rowborough Farm but also towards the west of the site in small numbers.	**Local**	Loss of potential nesting and foraging habitat through the loss of hedgerow sections and scrub and the loss of the arable landscape within the site.	Once mature, areas within the GI where scrub patches and ruderal weedy vegetation is allowed to establish are likely to continue to provide nesting and foraging opportunities for linnet. The site is likely to continue to provide nesting opportunities for a small number of breeding pairs particularly along the arable edge.	**Negligible**
Red Kite	Breed in small to medium sized deciduous woodland in rolling grassland and arable farmland areas. Both the native welsh and reintroduced programmes have done remarkably well and numbers are increasing.	Singletons observed to fly over the site only in the breeding and wintering seasons.	**Site**	Due to the lack of suitably established woodland habitat present within the site, no breeding habitat is available on site. The site provides some potential foraging opportunities for the small local population or individual passing birds.	None suggested	**Negligible**
Redwing	Widespread winter visitor in Britain and Ireland between October and March. Generally rather shy feeding in hedges and orchards and open areas of short grass. Visits farmland, parks and large gardens.	As with fieldfare, small to moderate numbers of redwing were observed throughout the site mainly within hedgerows.	**Local**	Loss of foraging habitat in the form of open pasture fields, hedgerows and scrub habitat.	Provision of new planting should include the provision of fruiting species which will provide further foraging habitat for moderate numbers of redwing.	**Negligible**

Species	Habitat Requirements	Species Account	Nature Conservation Value†	Characterisation of Unmitigated Impact	Suggested Possible Mitigation/ Compensation/ Enhancements	Residual Impact††
Reed Bunting	Traditional habitat is that of prolific fairly low vegetation, mainly associated with intense soil moisture. Increasingly found in cultivated drier habitats.	Breeding ï Restricted to habitats adjacent to the River Cole corridor, and mainly outside of the application boundary. Wintering ï Greater number of birds recorded than in the breeding season often foraging on seed bearing herbs. Still restricted to the eastern section of the site.	**Local**	Some minor loss of foraging habitats at the east of the site, particularly in the winter where birds were observed within the application boundary.	Through the provision of small wetland features throughout the site including SuDS features, reed bunting is likely to become more widespread across the site in similar numbers to those recorded in recent surveys.	**Negligible**
Skylark	Ground nesting birds favouring open farmland habitats where short, grassy or sparse vegetation provides nesting cover and foraging opportunities. Plant and animal material taken at all times of the year, but insects especially important in summer, cereal grain and weed seeds in autumn, leaves and weed seeds in winter, and cereal grain in spring.	Breeding ï Given the size of the site, relatively small numbers of skylark were recorded and mainly restricted to those larger arable fields within the centre of the site. Wintering ï Small to moderate sized flocks recorded within arable field compartments, particularly containing stubble, which provides good foraging opportunities.	**Local**	Given the loss of arable habitats from site it is considered that they will inevitably be lost from the site as a potential breeding and wintering species.	None suggested, however it is possible that discreet areas of the green infrastructure if free from disturbance and managed appropriately may provide suitable habitat for skylark both in the winter and breeding season. It is however realised that numbers of skylark the site supports is highly likely to be significantly reduced especially from those numbers recorded in winter.	**Negative**

Species	Habitat Requirements	Species Account	Nature Conservation Value†	Characterisation of Unmitigated Impact	Suggested Possible Mitigation/ Compensation/ Enhancements	Residual Impact††
Song Thrush	Requires a mixture of trees or bushes and open habitat. Can be found in human habitat such as parkland, hedgerows, roadsides or other areas that satisfy the above criteria. Forages on a wide variety of invertebrates, including snails, supplemented by fruit when seasonally available.	Recorded throughout the year in small numbers across the site in hedgerows and scrub.	**Local**	Loss of hedgerow sections leading to loss of potential nesting, foraging habitat and cover in winter.	New areas of planting will provide compensatory foraging and nesting habitat throughout the site. Newly created and enhanced wetland features will provide an important foraging resource.	**Negligible**
Starling	Foraging occurs on grassland, arable crops and grazed pastureland, for a mixture of plant and invertebrate food sources. Invertebrate uptake increases during breeding season. Breeds in cavities, either natural hollow trees or gaps in building structures.	Breeding ï As with house sparrow, starling were recorded to the western extent of the site associated with the residential boundary. Breeding was confirmed just outside of the application boundary at the Mercuer Hotel in South Marston. Wintering ï Much more widespread across the site and recorded in moderate numbers.	**Local**	Loss of some existing foraging habitat in the form of grazed grassland field compartments.	The inclusion of starling boxes within the built environment would present the species with much improved nesting opportunities. Areas of grassland and wetland habitats will continue to provide suitable foraging opportunities for the species.	**Negligible**

Species	Habitat Requirements	Species Account	Nature Conservation Value†	Characterisation of Unmitigated Impact	Suggested Possible Mitigation/ Compensation/ Enhancements	Residual Impact††
Yellow Wagtail	In breeding season traditionally occupies fringes of wetlands, but will breed on farmland in the UK. Feeds on invertebrates often in association with grazing cattle and sheep.	Breeding ï At least one pair confirmed breeding within the large central arable field (TN1, Figure 1) which supported a cereal crop at the time of the breeding bird surveys.	**Local**	Loss of arable habitat is likely to remove all potential nesting habitat from the site.	It is possible that discreet areas within the green infrastructure such as the SuDS features will provide some possible foraging opportunities to any birds within the surrounding arable landscape or to birds on passage. It is however most likely that yellow wagtail will be lost as a breeding species from the site.	**Negative**
Yellow-hammer	Traditionally based on edges of open areas of forest and fringing scrub of gorse, broom and hawthorn. Extends widely across cultivated land with hedges, plantations, and paths. Feeds mainly on grass seeds, invertebrates in the breeding season. Feeds wholly on the ground by hedges, tracks and newly sown fields.	Breeding ï A small number of pairs recorded and confirmed breeding centrally within the site. Wintering ï Moderate sized flocks recorded on each survey occasion particularly where stubble fields persisted which provided good foraging habitat.	**Local**	Loss of hedgerow sections and arable farmland is likely to significantly reduce the nesting and foraging opportunities to yellowhammer.	Some features of the green infrastructure proposals where habitat management is less formal may provide some foraging opportunities to surrounding populations. Although boundary features may continue to provide some limited nesting opportunities, the value of the developed site to yellowhammer is likely to be significantly reduced.	**Negative**

Ä Based **upon criteria set out in Table 3 and professional judgement.**

ÄÄ Assumes that **any suggested or proposed mitigation, compensation or enhancements are undertaken in full.**

Impacts of Habitat Loss on Other Species

5.4 It is identified that through the loss of the out buildings at Rowborough Farm, swallow (BoCC amber list, LBAP) is likely to lose potential nesting opportunities from the site. However, given the provision of areas of grassland and wetland habitat throughout the green infrastructure, the local swallow population is likely to benefit from enhanced foraging habitat leading to negligible residual impacts. Such features will also provide suitable foraging opportunities for swift (BoCC amber list, LBAP) and house martin (BoCC amber list). The provision of suitable swift and house martin nest boxes throughout the built environment will lead to positive residual impacts on these two species.

5.5 Impacts from the loss of hedgerows are also identified on the range of common and widespread species including a range thrushes, finches and warblers. The loss of hedgerow sections has the potential to remove nesting habitat and winter forage and cover. The provision of further structural planting within the extensive proposed green infrastructure including the planting of new hedgerows will in most part compensate the loss of hedgerows and residual impacts on the more generalist species on habitat loss is considered negligible.

Disturbance Impacts

5.6 Construction operations have the potential to disturb birds using the site for roosting, foraging, and breeding. Operations which will disturb breeding birds include noise from vegetation clearance, initial ground works and some construction activities such as piling, which are of low frequency but of high amplitude. Active, high level, infrequent disturbance causes most birds to be displaced for short periods (Treweek 1999). During the breeding season disturbance may lead to nest desertion or the avoidance of the area and reduce the suitability of retained nesting areas such as the hedgerows. Where necessary, the use of buffer zones around any active nests will minimise potential effects. Ground clearance works across the site are likely to be phased and will follow the phasing of the development, both staggering removal and allowing for some habituation of breeding birds, minimising effects from the outset. Whilst there is some potential for breeding success to be reduced, this is not expected to affect the local conservation status on the majority of the bird species using the site for breeding. Disturbance effects on the majority of birds are therefore expected to be short-term and temporary and are significant at a site level only.

Bird Predation

5.7 The increase in domestic animals during the operational phase, particularly cat, may lead to an effect on small bird populations. Recent research is inconclusive as to the actual effect that domestic cats can have on wild populations, however, a precautionary approach is advised (Gibbons *et al* 2007). In some cases the warning provided by the use of collar-mounted devices, such as bells, has been shown to significantly reduce the success of bird predation by cats (Nelson *et al* 2005). It is

considered probable that in the absence of mitigation development will result in a minor site level impact to the breeding bird assemblage.

Mitigation

5.8 To avoid disturbance to breeding birds, ground clearance works and vegetation will be undertaken prior to the bird-breeding season (March to September inclusive). If this is not possible, vegetation will be checked prior to removal by an experienced ecologist. If active nests are found, vegetation will be left untouched and suitably buffered from works until all birds have fledged. Specific advice will be provided prior to undertaking the clearance. This would be a statutory requirement due to the protection of all nesting birds and their nests under the Wildlife and Countryside Act, 1981. A suitably qualified ecologist would supervise this.

5.9 Barn owl are afforded special protection under Schedule 1 of the Wildlife and Countryside Act 1981 (as amended). Under this legislation and in addition to the actions described in paragraph 2.1 it is an offence to intentionally or recklessly disturb a bird listed under Schedule 1 while it is building a nest or is in, on or near a nest containing eggs or young or to disturb the dependent young of such a bird. Construction works within the vicinity of the identified barn owl boxes will also be undertaken outside of the breeding bird season. If this is not possible, the tree will be checked to confirm the presence/absence of barn owl. Should barn owls be using the potential roost site during the breeding season a disturbance-free buffer zone around the tree will be maintained as to protect nesting barn owls from disturbance.

Other Enhancements

5.10 Additional enhancements that could be integrated with the ongoing management of the site include the erection of a mixture of nest box types in addition to those already mentioned (house sparrow terraces, swift, house martin and starling boxes). The following provides details of other suitable nest box types to be included within the scheme:

- A mixture of small hole (26mm and 32mm) boxes placed throughout the site on suitable trees and buildings will provide nesting opportunities for blue tit and great tit. These boxes generally have a high uptake rate.
- Small open fronted nest boxes again should be placed throughout the site especially on trees which support a climber such as ivy which provides a degree of concealment. These boxes typically attract robin, blackbird and spotted flycatcher *Muscicapa striata*.
- A mixture of more specialised nest boxes should be placed within the more established boundary habitats and should include boxes for nuthatch, stock dove tawny owl *Strix aluco* and kestrel.

6.0 CONCLUSION

6.1 The results of the breeding and wintering bird surveys show that the site supports a species assemblage consisting in the most part of widespread, common and abundant species associated with arable farmland and woodland edge (scrub, hedgerow) habitats. Discreet areas of more established semi-natural habitats including the River Cole, the South Marston Brook and associated areas (habitats which in most part are to be retained) provide some opportunities for occasional use by more localised and uncommon to scarce species. No significant populations of any notable species were recorded, although it was recognised that the site supported moderate flocks of farmland birds during the winter, for which negative residual impacts have been recognised.

6.2 Given the presence of a number of NERC S41 and BoCC red and amber listed species and only the occasional use of the site by more specialised species, the overall bird assemblage is considered to be of Local nature conservation value.

6.3 It is recognised that the species assemblage is likely to change in composition and diversity with the proposed development benefiting birds of more urban environs. In addition, habitat creation throughout the green infrastructure including areas of wetland, tree, and shrub planting is likely to provide further opportunities for other notable species currently absent from the site. The resultant bird assemblage that develops is likely to equivalent in value to that recorded.

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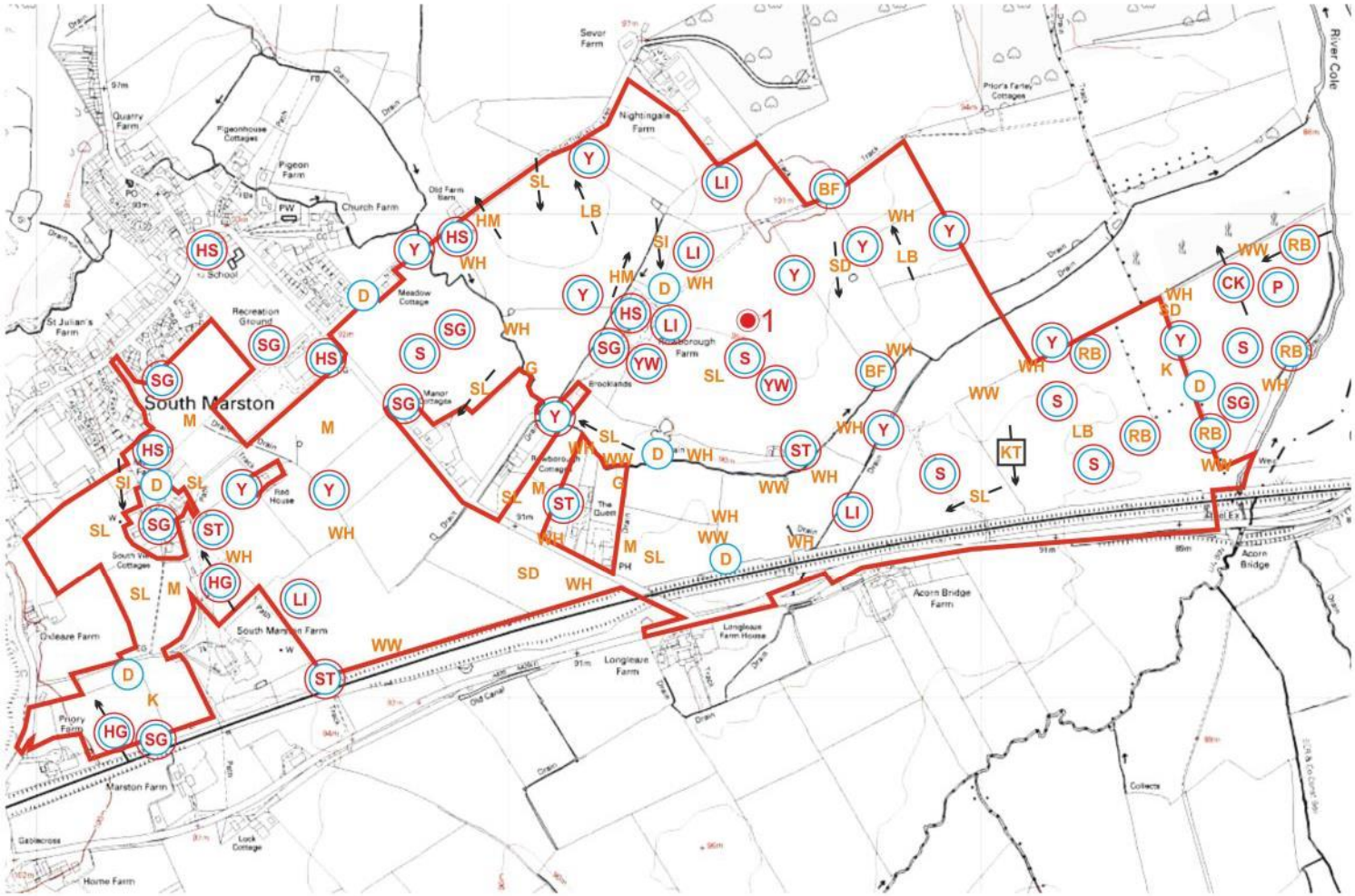

BoCC Red Listed Species

CK	CUCKOO	S	SKYLARK
P	GREY PARTRIDGE	ST	SONG THRUSH
HG	HERRING GULL	SG	STARLING
HS	HOUSE SPARROW	Y	YELLOWHAMMER
LI	LINNET	YW	YELLOW WAGTAIL

BoCC Amber Listed Species

BF	BULLFINCH	KT	RED KITE
D	DUNNOCK	RB	REED BUNTING
G	GREEN WOODPECKER	SD	STOCK DOVE
HM	HOUSE MARTIN	SL	SWALLOW
K	KESTREL	SI	SWIFT
LB	LESSER BLACK-BACKED GULL	WH	WHITETHROAT
M	MISTLE THRUSH	WW	WILLOW WARBLER

- 1 TARGET NOTE
- WCA SCHEDULE 1
- PRIORITY SPECIES (NERC S.41)
- LOCAL BAP
- SPECIES ONLY OBSERVED TO FLY OVER THE SITE AT THIS LOCATION
- SURVEY BOUNDARY

Hallam Land Management, Taylor Wimpey &Hannick Homes

EASTERN VILLAGES, SWINDON

BREEDING BIRD SURVEY 2011

NTS @ A3 TJM/PH 14.10.2013

N

Figure 1

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masterplanning environmental assessment landscape design urban design ecology architecture arboriculture

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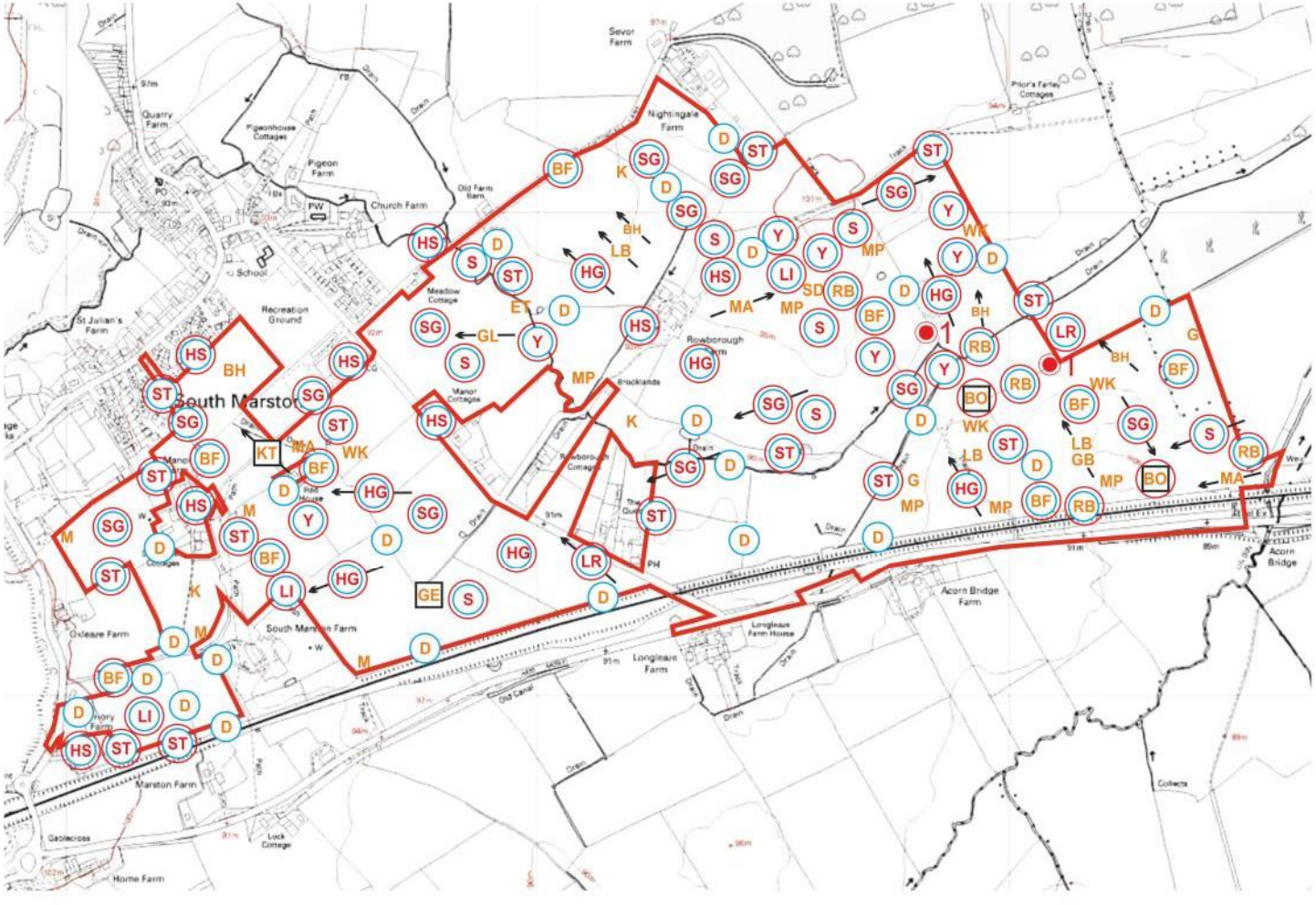

BoCC Red Listed Species

HG	HERRING GULL	S	SKYLARK
HS	HOUSE SPARROW	ST	SONG THRUSH
LI	LINNET	SG	STARLING
LR	LESSER REDPOLL	Y	YELLOWHAMMER

BoCC Amber Listed Species

BO	BARN OWL	KT	RED KITE
BF	BULLFINCH	LB	LESSER BLACK-BACKED GULL
BH	BLACK-HEADED GULL	ET	LITTLE EGRET
D	DUNNOCK	M	MISTLE THRUSH
G	GREEN WOODPECKER	MA	MALLARD
GE	GREEN SANDPIPER	MP	MEADOW PIPIT
GB	GREAT BLACK-BACKED GULL	RB	REED BUNTING
GL	GREY WAGTAIL	SD	STOCK DOVE
K	KESTREL	WK	WOODCOCK

WCA SCHEDULE 1

PRIORITY SPECIES (NERC S.41)

LOCAL BAP

SPECIES ONLY OBSERVED TO FLY OVER THE SITE AT THIS LOCATION

SURVEY BOUNDARY

1 TARGET NOTE

Hallam Land Management, Taylor Wimpey &Hannick Homes

EASTERN VILLAGES, SWINDON

WINTER BIRD SURVEY 2012-13

NTS @ A3 DJC/SLS 14.10.2013

N

Figure 2

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APPENDIX A: BREEDING BIRD SURVEY RESULTS

Species	Latin	Survey 1 12.05.11	Survey 2 01.06.11	Survey 3 21.06.11	Conservation Status	Breeding status
Blackbird	*Turdus merula*	15 + 2 juveniles	38	28+ 2 juveniles	Green list	Confirmed
Blackcap	*Sylvia atricapilla*	6	7	7	Green list	Probable
Blue tit	*Cyanistes caeruleus*	9	12 + 2 families	13	Green list	Confirmed
Bullfinch	*Pyrrhula pyrrhula*	2	1	2	Amber list NERC Act (S41) LBAP	Probable
Buzzard	*Buteo buteo*	2	3	1	Green list	Possible
Carrion crow	*Corvus corone*	16	15	14	Green list	Probable
Chaffinch	*Fringilla coelebs*	14	19	6	Green list	Confirmed
Chiffchaff	*Phylloscopus collybita*	13	9	6	Green list	Probable
Coal tit	*Parus ater*	1	0	0	Green list	Possible
Collared dove	*Streptopelia decaocto*	5	6	7	Green list	Probable
Cuckoo	*Cuculus canorus*	1	0	0	Red list NERC Act (S41) LBAP	Possible
Dunnock	*Prunella modularis*	10	9	10	Amber list NERC Act (S41)	Probable
Feral pigeon	*Columba livia*	35	1	0	Green list	Possible
Garden warbler	*Sylvia borin*	3	1	1	Green list	Probable
Goldcrest	*Regulus regulus*	1	1	1	Green list	Possible
Goldfinch	*Carduelis carduelis*	7	38	19 + 2 families	Green list	Confirmed
Great spotted woodpecker	*Dendrocopos major*	0	0	1	Green list	Possible
Great tit	*Parus major*	4	14 including juveniles + 4 families	3 + 1 family	Green list	Confirmed
Green woodpecker	*Picus viridis*	1	1	0	Amber list	Possible
Greenfinch	*Carduelis chloris*	28	14	8	Green list	Probable
Grey heron	*Ardea cinerea*	0	1	2	Green list	Non-breeder

Species	Latin	Survey 1 12.05.11	Survey 2 01.06.11	Survey 3 21.06.11	Conservation Status	Breeding status
Grey partridge	*Perdix perdix*	2	0	0	Red list NERC Act (S41) LBAP	Possible
Herring gull	*Larus argentatus*	1	2	0	Red list NERC Act (S41) LBAP	Non-breeder
House martin	*Delichon urbica*	0	24	6	Amber list LBAP	Possible
House sparrow	*Passer domesticus*	5 colonies	4 colonies	4 colonies	Red list NERC Act (S41) LBAP	Probable
Jackdaw	*Corvus monedula*	85	62	14	Green list	Probable
Jay	*Garrulus glandarius*	0	1	0	Green list	Possible
Kestrel	*Falco tinnunculus*	0	1	1	Amber list	Possible
Lesser Black-backed Gull	*Larus fuscus*	0	1	3	Amber list	Non-breeder
Lesser whitethroat	*Sylvia curruca*	3	3	5	Green list	Probable
Linnet	*Carduelis cannabina*	11	8	5	Red list NERC Act (S41) LBAP	Probable
Long-tailed Tit	*Aegithalos caudatus*	3 including juveniles	6	1 family	Green list	Confirmed
Magpie	*Pica pica*	12	12	10	Green list	Probable
Mistle thrush	*Turdus viscivorus*	4	0	2	Amber list	Probable
Moorhen	*Gallinula chloropus*	2	0	0	Green list	Possible
Pheasant	*Phasianus colchicus*	1	2	3	Not listed	Possible
Pied wagtail	*Motacilla alba*	0	8 including juveniles	1	Green list	Confirmed
Red kite	*Milvus milvus*	0	0	1	Schedule 1 Amber list	Non-breeder
Red-legged Partridge	*Alectoris rufa*	1	0	2	Not listed	Possible
Reed bunting	*Emberiza schoeniclus*	4	0	5	Amber list NERC Act (S41) LBAP	Probable
Robin	*Erithacus rubecula*	15	17 + 1 juvenile	10	Green list	Confirmed
Rook	*Corvus frugilegus*	41	20	0	Green list	Probable
Sedge warbler	*Acrocephalus schoenobaenus*	1	0	0	Green list	Possible

Species	Latin	Survey 1 12.05.11	Survey 2 01.06.11	Survey 3 21.06.11	Conservation Status	Breeding status
Skylark	*Alauda arvensis*	2	5	5	Red list NERC Act (S41) LBAP	Probable
Song thrush	*Turdus philomelos*	2	5	2	Red list NERC Act (S41) LBAP	Probable
Sparrowhawk	*Accipiter nisus*	0	0	1	Green list	Possible
Starling	*Sturnus vulgaris*	30 including juveniles	30	55	Red list NERC Act (S41) LBAP	Probable
Stock dove	*Columba oenas*	5	1	7	Amber list	Probable
Swallow	*Hirundo rustica*	19	47	14+ 1 family	Amber list LBAP	Confirmed
Swift	*Apus apus*	0	5	2	Amber list LBAP	Non-breeder
Whitethroat	*Sylvia communis*	17	24 + 1 family	9 + 1 family	Amber list	Confirmed
Willow warbler	*Phylloscopus trochilus*	9	15	2	Amber list	Probable
Woodpigeon	*Columba palumbus*	64	39	83	Green list	Confirmed
Wren	*Troglodytes troglodytes*	12	19	13	Green list	Probable
Yellow wagtail	*Motacilla flava*	4	1	2	Red list NERC Act (S41) LBAP	Confirmed
Yellowhammer	*Emberiza citronella*	9	9	7 + 1 pair	Red list NERC Act (S41) LBAP	Confirmed
Total No. Species	**56**	**49**	**47**	**48**		

Breeding Status evidence can be broken down into four sections, each with their own codes:

Confirmed breeder
DD – distraction display or injury feigning
UN – used nest or eggshells found from this season
FL – recently fledged young or downy young
ON – adults entering or leaving nest-site in circumstances indicating occupied nest
FF – adult carrying faecal sac or food for young
NE – nest containing eggs
NY – nest with young seen or heard

Probable breeder - Evidence accumulated during the survey indicates that the bird species is breeding on site.
P – pair in suitable nesting habitat
T – permanent territory (defended over at least 2 survey occasions)
D – courtship and display
N – visiting probable nest site
A – agitated behaviour

I – brood patch of incubating bird (from bird in hand)
B – nest building or excavating nest-hole

Possible breeder - Evidence accumulated during the survey indicates that the bird species could be breeding on site, but the evidence is less conclusive than that obtained for probable breeders.
H – observed in suitable nesting habitat
S – singing male

Non-breeder
F – flying over
M – migrant
U – summering non-breeder
UH – observed in unsuitable nesting habitat

APPENDIX B: WINTERING BIRD SURVEY RESULTS

Species	Scientific Name	Survey 1 18.12.12	Survey 2 10.01.13	Survey 3 04.02.13	Conservation Status
Barn Owl	*Tyto alba*	0	0	1	Schedule 1 Amber list LBAP
Blackbird	*Turdus merula*	19	62	42	Green list
Black-headed gull	*Chroicocephalus ridibundus*	56	0	3	Amber list
Blue tit	*Parus caeruleus*	29	28	35	Green list
Bullfinch	*Pyrrhula pyrrhula*	6	5	1	NERC S41 Amber list LBAP
Buzzard	*Buteo buteo*	2	3	3	Green list
Carrion crow	*Corvus corone*	36	23	14	Green list
Chaffinch	*Fringilla coelebs*	64	148	109	Green list
Chiffchaff	*Phylloscopus colybita*	1	0	0	Green list
Collared dove	*Streptopelia decaocto*	4	10	14	Green list
Dunnock	*Prunella modularis*	13	18	6	NERC S41 Amber list
Feral pigeon	*Columba livia*	30	16	4	Green list
Fieldfare	*Turdus pilaris*	47	68	37	Schedule 1 Red list
Goldcrest	*Regulus regulus*	4	4	3	Green list
Goldfinch	*Carduelis carduelis*	14	95	6	Green list
Great black-backed gull	*Larus marinus*	2	0	0	Amber list
Great spotted woodpecker	*Dendrocopos major*	2	3	1	Green list
Great tit	*Parus major*	14	15	15	Green list
Green Sandpiper	*Tringa ochropus*	0	0	1	Schedule 1 Amber List
Green woodpecker	*Picus viridis*	0	2	0	Amber list
Greenfinch	*Carduelis chloris*	5	1	1	Green list

Species	Scientific Name	Survey 1 18.12.12	Survey 2 10.01.13	Survey 3 04.02.13	Conservation Status
Grey Wagtail	*Motacilla cinerea*	0	1	0	Amber List
Herring Gull	*Larus argentatus*	c.70	8	4	NERC S41 Red list LBAP
House sparrow	*Passer domesticus*	5xCol (+2)	3xCol (+2)	3xCol (+2)	NERC S41 Red list LBAP
Jackdaw	*Corvus monedula*	53	60	172	Green list
Jay	*Garrulus glandarius*	1	0	2	Green list
Kestrel	*Falco tinnunculus*	2	0	0	Amber list
Lesser black-backed gull	*Larus fuscus*	c.190	24	2	Amber list
Lesser Redpoll	*Carduelis cabaret*	4	0	0	NERC S41 Red list LBAP
Linnet	*Carduelis cannabina*	6	3	6	NERC S41 Red list LBAP
Little Egret	*Egretta garzetta*	1	0	0	Amber list
Long-tailed tit	*Aegithalos caudatus*	4 flocks	2 flocks	3 flocks	Green list
Magpie	*Pica pica*	13	12	15	Green list
Mallard	*Anas platyrhynchos*	2	7	3	Amber list
Meadow Pipit	*Anthus pratensis*	3	6	6	Amber list
Mistle thrush	*Turdus viscivorus*	2	5	0	Amber list
Moorhen	*Gallinula chlorpus*	0	0	1	Green list
Pheasant	*Phasianus colchicus*	6	8	3	Introduced
Pied wagtail	*Motacilla alba*	4	8	2	Green list
Red-legged partridge	*Alectoris rufa*	6	4	8	Introduced
Red Kite	*Milvus milvus*	0	1	0	Schedule 1 Amber list
Redwing	*Turdus iliacus*	86	153	11	Schedule 1 Red list

Species	Scientific Name	Survey 1 18.12.12	Survey 2 10.01.13	Survey 3 04.02.13	Conservation Status
Reed bunting	*Emberiza schoeniclus*	3	12	10	NERC S41 Amber list LBAP
Robin	*Erithacus rubecula*	14	22	18	Green list
Rook	*Corvus frugilegus*	12	26	70	Green list
Siskin	*Carduelis spinus*	0	1	0	Green list
Skylark	*Alauda arvensis*	19	33	8	NERC 41 Red list LBAP
Song thrush	*Turdus philomelos*	7	9	4	NERC S41 Red list LBAP
Starling	*Sturnus vulgaris*	70	189	2	NERC S41 Red list LBAP
Stock dove	*Columba oenas*	1	0	0	Amber list
Woodcock	*Scolopax rusticola*	3	1	1	Amber list
Woodpigeon	*Columba palumbus*	316	166	296	Green list
Wren	*Troglodytes troglodytes*	6	9	9	Green list
Yellowhammer	*Emberiza citrinella*	42	41	21	NERC S41 Red list LBAP
Total Species	**54**	**47**	**42**	**43**	